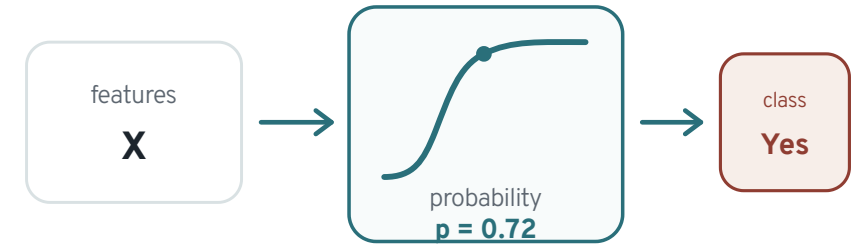


LOGISTIC REGRESSION AND CLASSIFICATION



Prof. Rebin Muhammad • ISLP Chapter 4

From **probabilities** to **classification decisions**

CLASSIFICATION ASSIGNS OBSERVATIONS TO CLASSES

CLASSIFICATION

The response variable is **categorical**.

Examples: **Default / No Default**, Spam / Not Spam.

REGRESSION

The response variable is **quantitative**.

Examples: credit-card balance, income, house price.

BINARY VS. MULTICLASS

Binary classification: two possible classes.

Multiclass classification: more than two classes.

This week, we focus on the binary case: **Yes / No**.

PROBABILITY IS NOT A CLASS

A classifier may first estimate a **probability**.

A **threshold** then converts that probability into a predicted class.

The goal is to generalize to new data, not memorize the training set. For logistic regression, the model estimates a probability; a decision rule converts that probability into a class.

CAN WE PREDICT WHO WILL DEFAULT?

A bank wants to predict whether a credit-card customer will **default**. We will use the **Default** data from ISLP Chapter 4 as our running example.

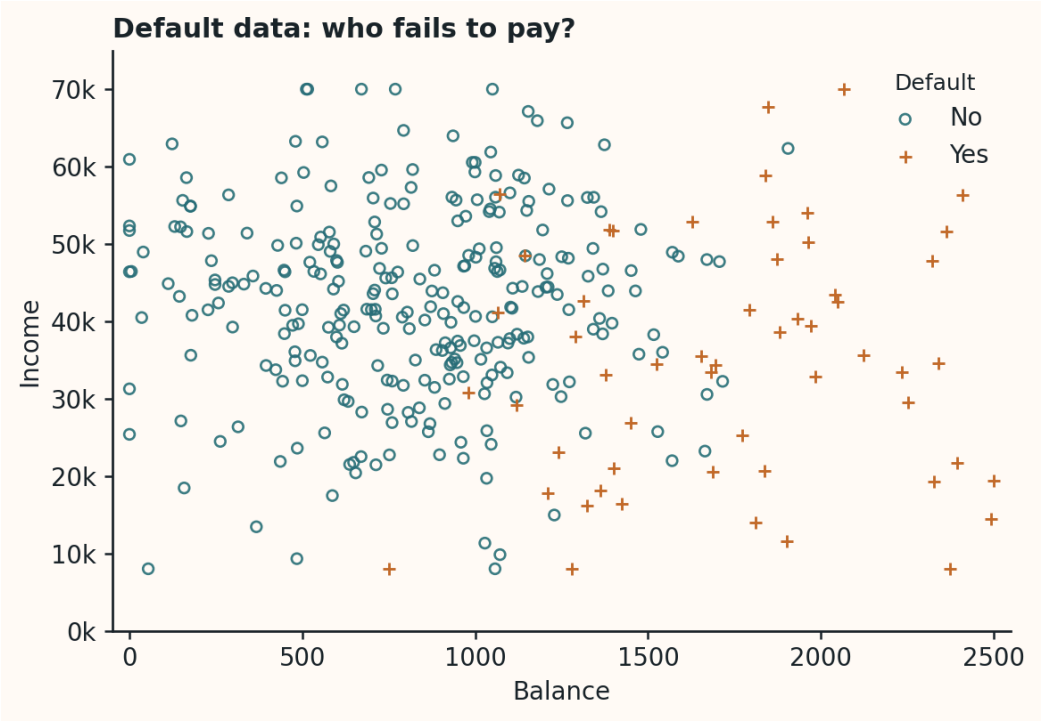
Example observations

Balance	Income	Student
\$800	\$42,000	No
\$1,950	\$38,000	Yes

Predictors, X: balance, income, student status

Target, Y: default – Yes or No

Because the training data contain both the predictors and the known outcome, this is **supervised classification**.



What does logistic regression learn from these labeled examples? It does not jump directly from the predictors to Yes or No.

LOGISTIC REGRESSION PREDICTS A PROBABILITY FIRST



Look at this customer

Balance = **\$1,850**

Income = **\$45,000**

Student = **Yes**

Suppose the fitted model estimates

$$P(\text{Default} = \text{Yes} \mid X) = 0.72$$

WHAT DOES 0.72 MEAN?

Given this customer's predictors, the model estimates a **72% probability of default**.

It does **not** mean: “this customer will definitely default.”

If the classification threshold is **0.50**:

$$0.72 \geq 0.50 \rightarrow \text{predicted class} = \text{Default} = \text{Yes}$$

Probability and predicted class are not the same thing. Logistic regression estimates the probability first; the threshold determines the class.

START WITH A LINEAR COMBINATION OF THE PREDICTORS

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p$$

Each predictor X_j is multiplied by a coefficient β_j . The model learns these coefficients from the training data.

We call z the **linear predictor**. It summarizes the predictors into one number, but it is **not yet a probability**.

THE PROBLEM

The linear predictor can be **any real number**.

-4.2 0.8 7.1

But a probability must stay between **0 and 1**.

SO WE CANNOT USE Z DIRECTLY

If we tried to interpret z as $P(Y = 1 \mid X)$, we could get values such as:

1.4 or **-0.3**

Those cannot be probabilities.

What do we need? A function that takes any value of $z \in (-\infty, \infty)$ and maps it to a probability in $(0, 1)$.

Next: the **logistic function** does exactly this.

THE LOGISTIC FUNCTION

TURN THE LINEAR PREDICTOR INTO A PROBABILITY

From the previous step, we have a linear predictor

$$z = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

The **logistic function** transforms z into a valid probability:

$$p(X) = \frac{e^z}{1+e^z} = \frac{1}{1+e^{-z}}$$

The result:

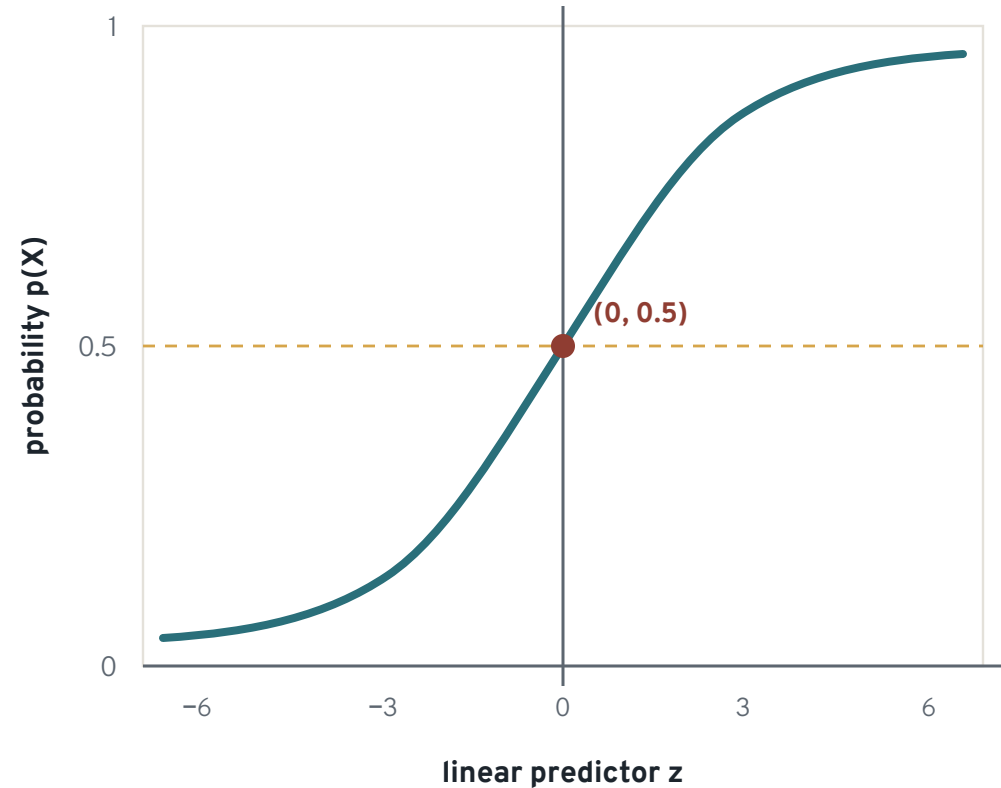
Any value of z is mapped into $0 < p(X) < 1$.

Here, $p(X) = P(Y = 1 | X)$.

Important: when $z = 0$, $p(X) = 0.50$.

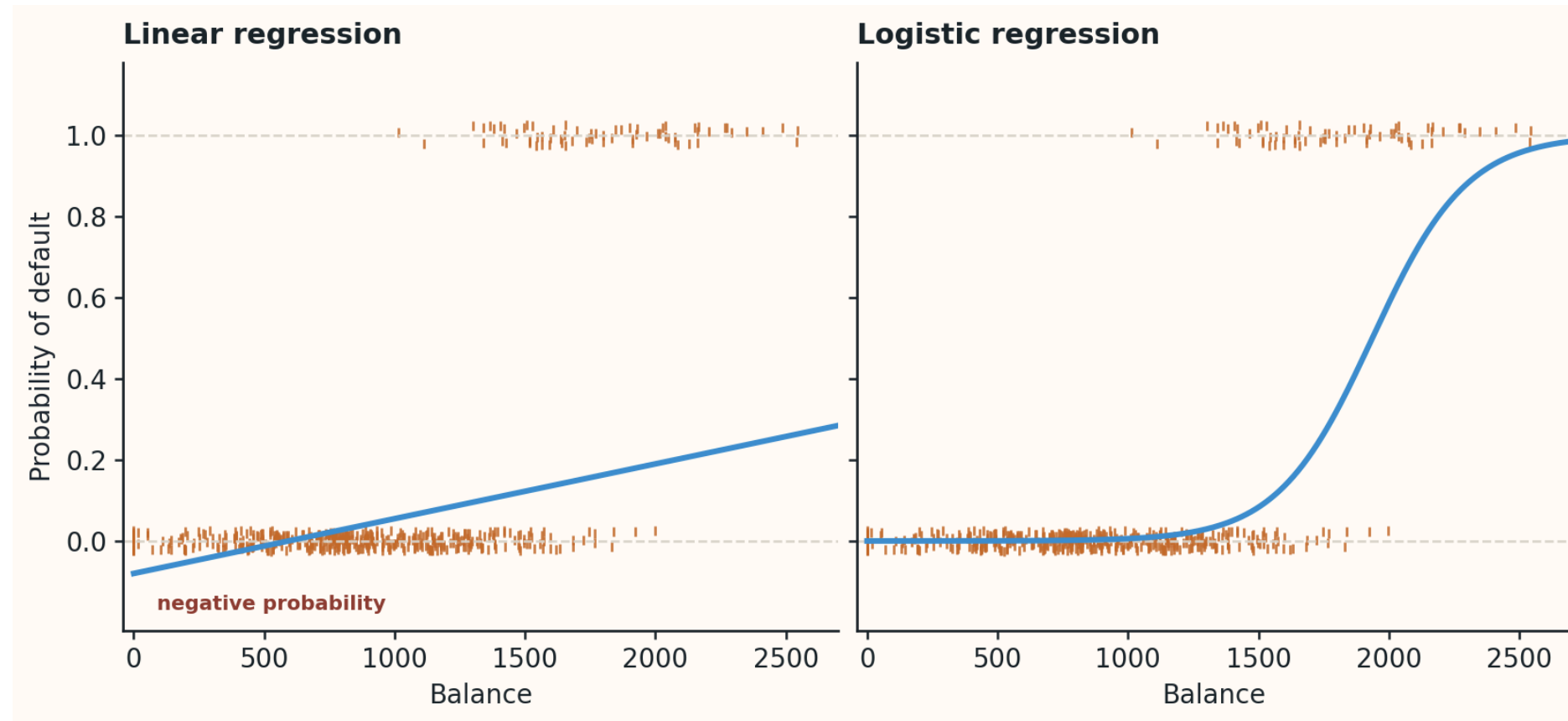
Negative $z \rightarrow$ probability below 0.50

Positive $z \rightarrow$ probability above 0.50



The S-shape is the key: z can range from $-\infty$ to $+\infty$, while the predicted probability always stays between 0 and 1.

WHY LOGISTIC REGRESSION INSTEAD OF A STRAIGHT LINE?



Linear regression

A straight line is unrestricted, so fitted values can fall **below 0 or above 1**.

Logistic regression

The S-shaped curve keeps every estimated probability **between 0 and 1**.

Key idea: for a binary outcome, we want to model $P(Y = 1 \mid X)$ — and a probability must remain between 0 and 1.

LOG-ODDS, OR THE LOGIT

$$\log \left(\frac{p(X)}{1-p(X)} \right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p = z$$

The logistic model is **linear in the log-odds**, not linear in the probability itself.

ODDS

The quantity $\frac{p}{1-p}$ is called the **odds**.

If $p = 0.20$:

$$\text{odds} = \frac{0.20}{0.80} = 0.25 = 1 : 4$$

If $p = 0.90$:

$$\text{odds} = \frac{0.90}{0.10} = 9 = 9 : 1$$

LOG-ODDS

Taking the logarithm of the odds gives the **logit**:

$$\log \left(\frac{p}{1-p} \right) = z$$

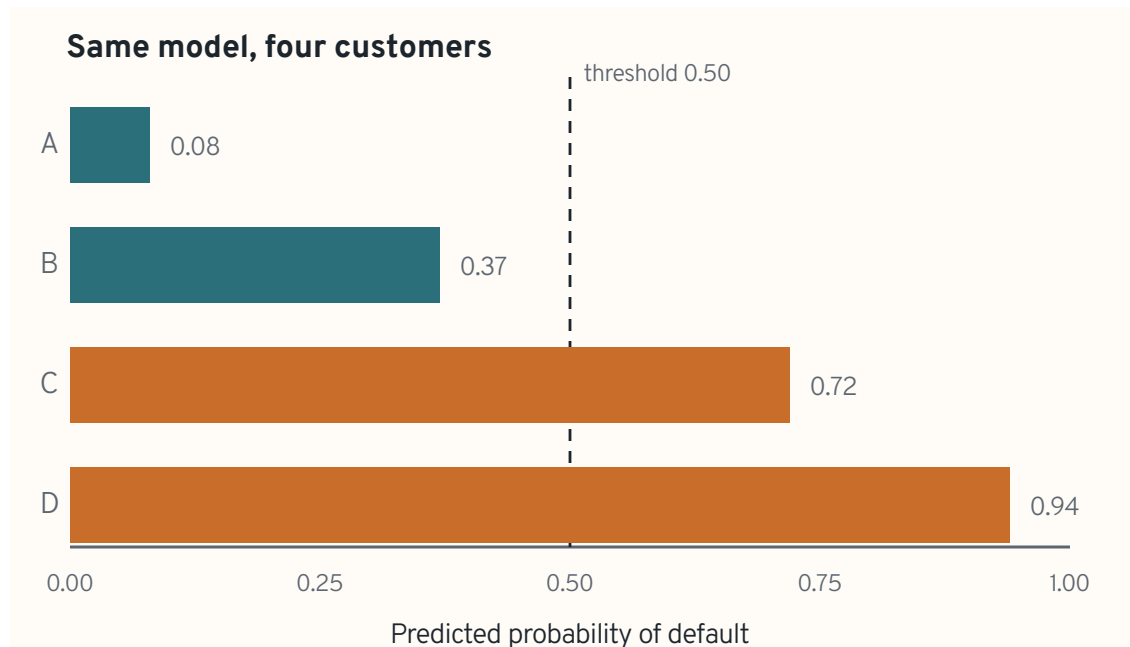
So the **linear predictor** z is exactly the model's **log-odds**.

How do we interpret β_j ? Holding the other predictors fixed, a one-unit increase in X_j changes the **log-odds** by β_j .

Equivalently, e^{β_j} is the factor by which the **odds** are multiplied.

The connection: predictors $X \rightarrow$ linear predictor $z = \text{log-odds} \rightarrow$ probability $p(X)$.

PROBABILITIES ARE OFTEN THE MORE USEFUL OUTPUT



A CLASSIFIER DOES NOT HAVE TO OUTPUT ONLY A LABEL

A binary classifier can return an estimated probability of **Default = Yes**.

That probability contains more information than a simple Yes / No prediction.

For example, **0.94** and **0.51** would both be classified as “Yes” using a 0.50 threshold, but they are very different probability estimates.

Customer A: very low predicted probability of default.

Customer D: very high predicted probability of default.

B and C: less extreme predictions. The threshold converts each probability into a class.

Keep the probabilities when they matter. Converting a probability to a class throws away information about how far the prediction is from the decision threshold.

A THRESHOLD TURNS A PROBABILITY INTO A CLASS

Suppose the positive class is **Default = Yes**. A common threshold is **0.50**:

$$\hat{Y} = \begin{cases} 1 \text{ (Default = Yes)} & \text{if } p(X) \geq 0.50 \\ 0 \text{ (Default = No)} & \text{if } p(X) < 0.50 \end{cases}$$

Customer	P(Default)	Predicted class
A	0.08	No
B	0.37	No
C	0.72	YES
D	0.94	YES

0.50 IS COMMON, NOT REQUIRED

The threshold is a **decision rule**, not something fixed by logistic regression.

If missing a true default is especially costly, we may choose a **lower threshold**.

Watch Customer B:

$$p = 0.37$$

Threshold 0.50 → **No Default**

Threshold 0.30 → **Default**

Same probability. Same fitted model. **Different decision rule.**

Lowering the threshold generally predicts **more observations as positive**.

Separate the model from the decision rule: the model produces probabilities; the threshold converts those probabilities into predicted classes.

WHY MIGHT WE CHOOSE A THRESHOLD BELOW 0.50?

Consider a different example: a screening model estimates the probability that a patient has a serious disease.

Suppose a patient has $P(\text{disease}) = 0.37$.

At threshold **0.50**: $0.37 < 0.50 \rightarrow$ predicted **Negative**.

At threshold **0.20**: $0.37 \geq 0.20 \rightarrow$ predicted **Positive**.

Lowering the threshold predicts more cases as positive.

This generally means:

fewer false negatives (FN) but **more false positives (FP)**.

		PREDICTED	
		Negative	Positive
ACTUAL	Negative	TN true negative	FP false positive
	Positive	FN false negative	TP true positive

confusion matrix

Key idea

The model produces a **probability**. The threshold determines the **predicted class**.

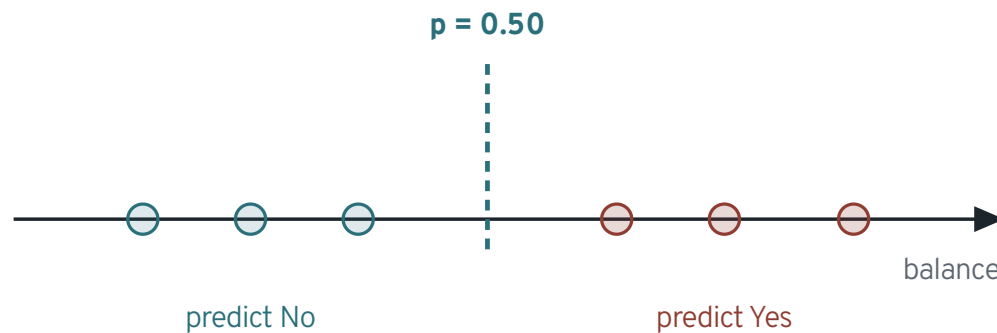
Which threshold is appropriate depends on the **cost of the two types of errors**.

Changing the threshold does **not** change the fitted model or its probabilities. It changes the decisions – and therefore the pattern of classification errors.

A CUTOFF IN ONE FEATURE BECOMES A LINE IN TWO

With one predictor, the threshold is a single cutoff point. With two predictors, the same 0.50 rule becomes a **decision boundary** in the plane.

One predictor



Key idea: one feature gives a cutoff point. Two features give a boundary line. With more predictors, we get a higher-dimensional boundary.

TWO PREDICTORS

Suppose we predict default using

$\mathbf{X}_1 = \text{Balance}$ $\mathbf{X}_2 = \text{Income}$

Then the linear predictor is

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

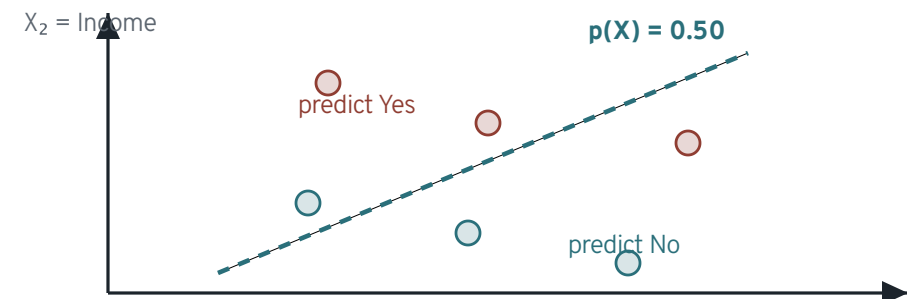
At a 0.50 threshold, we know that $p(\mathbf{X}) = 0.50$ exactly when $\mathbf{z} = 0$.

So the boundary is

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0$$

That equation describes a **line** in the (X_1, X_2) -plane.

Decision boundary in two predictors



The same threshold rule is still being used – it is just drawn in the space of the predictors instead of on a single number line.

AT A 0.50 THRESHOLD, THE DECISION BOUNDARY IS $Z = 0$

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0$$

With two predictors, this equation describes a **line**. The line separates observations predicted as class 0 from those predicted as class 1.

ONE SIDE OF THE BOUNDARY

$$z < 0 \rightarrow p(X) < 0.50$$

Predicted class: **0**

THE OTHER SIDE

$$z \geq 0 \rightarrow p(X) \geq 0.50$$

Predicted class: **1**

More predictors, same idea.

1 predictor $\rightarrow$ a **cutoff point**

2 predictors $\rightarrow$ a **line**

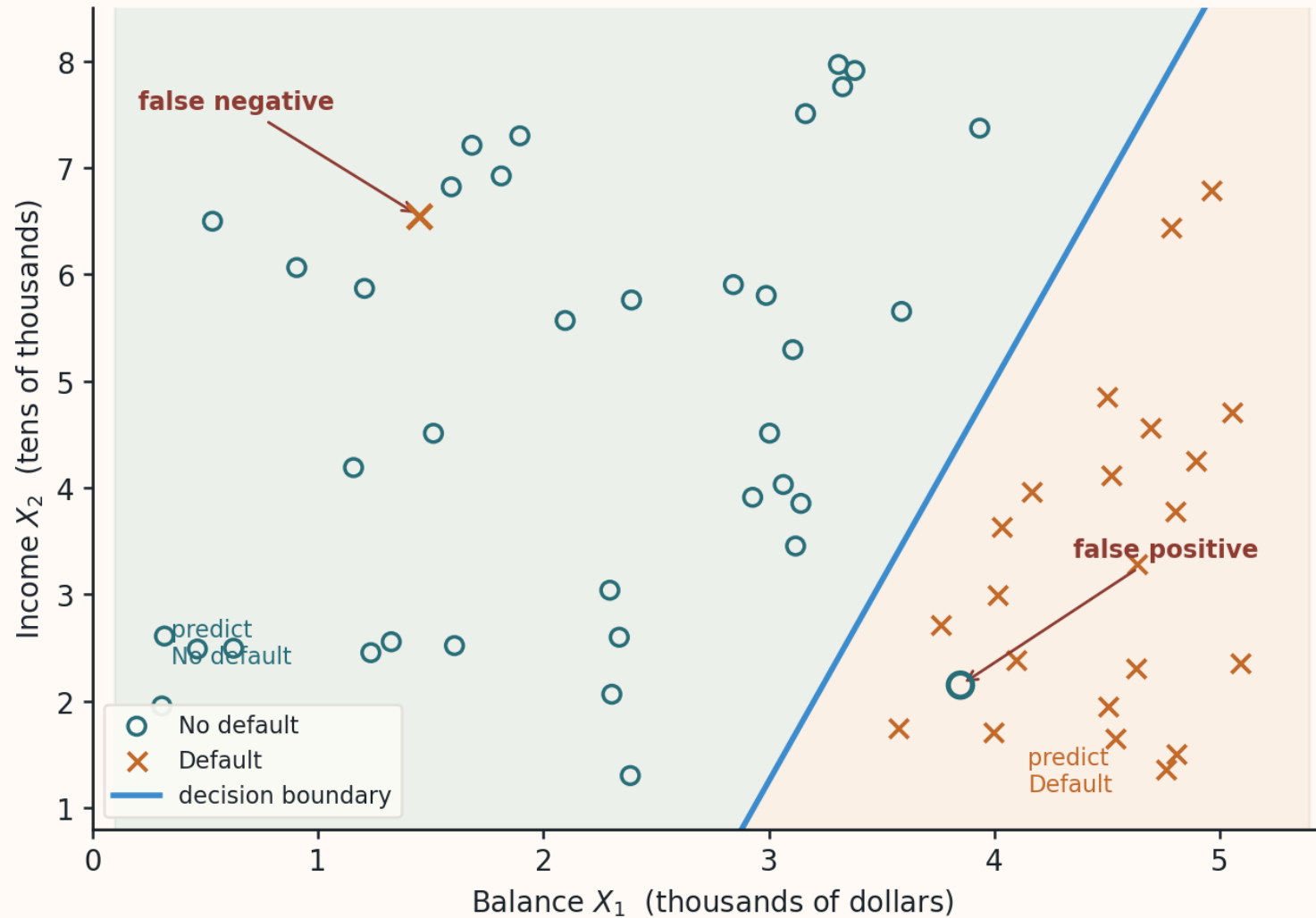
3 predictors $\rightarrow$ a **plane**

many predictors $\rightarrow$ a **hyperplane**

Remember: $z = 0$ is the boundary specifically for a **0.50 threshold**. Changing the probability threshold shifts the boundary.

Why “linear classifier”? The sigmoid is nonlinear, but the decision boundary is a linear equation in the predictors.

CAN YOU SEE THE TWO MISTAKES?



FALSE POSITIVE

Predicted Default, actually No. An orange region, teal point.

FALSE NEGATIVE

Predicted No, actually Default. A teal region, orange point.

A linear boundary cannot curve around every point. Some errors are the price of a simple rule.

A WORKED EXAMPLE

WOULD THESE CUSTOMERS DEFAULT?

Two features, scaled so the coefficients are readable:

X_1 : credit-card balance, in **\$1,000s**

X_2 : annual income, in **\$10,000s**

$$z = -4 + 1.5 X_1 - 0.4 X_2$$

At $p = 0.50$ we have $z = 0$, so the boundary is

$$-4 + 1.5 X_1 - 0.4 X_2 = 0$$

Customer A

Balance \$1,000 $\rightarrow X_1 = 1$

Income \$60,000 $\rightarrow X_2 = 6$

Customer B

Balance \$4,000 $\rightarrow X_1 = 4$

Income \$30,000 $\rightarrow X_2 = 3$

Higher balance pushes z up (more default risk). Higher income pushes z down.

FROM THE SCORE TO A CLASS

$$z = -4 + 1.5(1) - 0.4(6) = -4.9$$

Convert to a probability:

$$p(X) = \frac{e^{-4.9}}{1 + e^{-4.9}} \approx 0.007$$

$P(\text{Default}) \approx 0.7\%$. Since $0.007 < 0.50$, the model predicts
No default.

YOUR TURN: CUSTOMER B

$$X_1 = 4, \quad X_2 = 3$$

Compute z , then $p(X)$, then the class at 0.50.

$$z = -4 + 1.5(4) - 0.4(3) = ?$$

Answer

$$z = 0.8$$

$$p(X) \approx 0.69$$

Predicted class: Default = Yes

A large negative score is a tiny probability. The sigmoid is doing the squeezing.

HOW DOES THE COMPUTER LEARN THE COEFFICIENTS?

WHICH MODEL LOOKS BETTER?

Four customers have known outcomes. Two candidate models assign different probabilities of **Default = Yes**. We need a way to decide which model fits the observed data better.

Customer	Actual	Model A
A	No	0.05
B	No	0.15

Model A gives probabilities close to 0 for the No cases and close to 1 for the Yes cases.

Model B is only slightly above 0.50 for the Yes cases, and it is **confidently wrong** on customer B.

At a 0.50 threshold: **Model A gets 4/4 correct**, while **Model B gets 3/4 correct**.

But we want a method that uses the **probabilities themselves**, not only the final class labels.

Key idea

A good model assigns **high probability to the outcome that actually occurred**.

Next question: how can we turn these probabilities into a numerical measure of model fit?

FIND THE COEFFICIENTS THAT BEST FIT THE DATA

Loss measures how poorly the predicted probabilities match the observed outcomes.

Logistic regression chooses coefficients that **minimize log-loss** – equivalently, coefficients that **maximize the likelihood** of the observed data.

1. CHOOSE COEFFICIENTS

Start with values for $\beta_0, \beta_1, \dots, \beta_p$.

Use them to compute the **linear predictor** z and then the predicted probabilities.

2. MEASURE THE FIT

Compare the predicted probabilities with the observed **0 / 1 outcomes**.

Better probabilities → **lower log-loss**.
Poor or confidently wrong predictions → **larger loss**.

3. UPDATE AND REPEAT

Adjust the coefficients to improve the objective.

Repeat until further changes produce little or no improvement.

Gradient descent is one way to perform this search.

coefficients → probabilities → loss → update coefficients → repeat

The statistical goal is maximum likelihood. Numerical optimization is the computational tool used to find those coefficient estimates.

1. STATISTICAL INFERENCE

Logistic regression can be used to **understand relationships** between predictors and a binary outcome.

Here the main question is not just “Can we predict?” but “What does the fitted model tell us about the association?”

Example: Is credit-card balance associated with default?

If the model contains other predictors, we ask about the association with balance **while holding those other predictors fixed**.

For inference, we examine the estimated coefficients $\hat{\beta}_j$ together with:

standard errors • z -statistics • p -values • confidence intervals

The coefficient describes how the **log-odds** change with the predictor; e^{β_j} gives the corresponding multiplicative change in the odds.

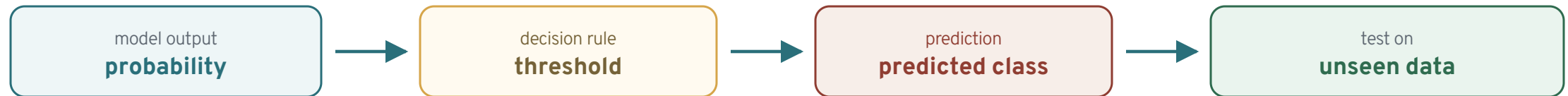
Typical inference question: Is there evidence that balance is associated with default, and how large is that association?

Inference: use the fitted model to study associations between predictors and the outcome.

2. MACHINE-LEARNING CLASSIFICATION

Now the goal is different: can the fitted model make useful predictions for **new customers it has never seen?**

The focus shifts from interpreting coefficients to **generalization** and predictive performance.



INFERENCE

What does the fitted model tell us about the **association** between predictors and the outcome?

Focus: coefficients, uncertainty, p -values, confidence intervals.

PREDICTION

How well does the fitted model perform on **new observations?**

Focus: predictive probabilities, classification errors, thresholds, and test performance.

Typical prediction question: Can balance, income, and student status help us predict whether a **new customer** will default?

Same logistic model – different goal. In DATA 201, we will emphasize prediction, classification, and generalization to unseen data.

TRY THIS

THRESHOLDS AND CLASSIFICATION ERRORS

Customer	Balance	P(Default)
A	\$400	0.05
B	\$1,100	0.32

1. Using a threshold of **0.50**, assign each customer a predicted class: **Default if $p \geq 0.50$** and **No Default if $p < 0.50$** . Which predictions are incorrect?

Answer

A, B, C → **No Default**

D, E, F → **Default**

C: False Negative (FN)

D: False Positive (FP)

2. Now lower the threshold to **0.30**. Which customers change predicted class? How do the numbers of **false positives** and **false negatives** change?

Answer

B and C change predicted class.

Threshold 0.50 → **1 FP + 1 FN**

Threshold 0.30 → **2 FP + 0 FN**

Is 0.30 necessarily better?

No. Lowering the threshold eliminates the false negative here, but creates an additional false positive.

The best threshold depends on the consequences of each type of error.

PROBABILITY, INFERENCE, AND PREDICTION

3. Customer C actually defaults, but the model gives $P(\text{Default}) = 0.46$. Is the probability itself necessarily unreasonable?

No. A probability of 0.46 does not mean that Customer C cannot default. It means the model estimated a **46% chance of default**. The probability and the final class are different things. The **classification error** occurs after we impose the 0.50 threshold.

4. Which question is about **inference**, and which is about **prediction**?

Inference

“Does balance have a statistically significant relationship with default?”

Goal: understand and explain the relationship between variables.

Prediction

“Can balance help us accurately predict default for new customers?”

Goal: predict outcomes for new observations.

5. What is logistic regression trying to accomplish during training?

It chooses coefficients that make the observed outcomes as likely as possible – equivalently, it **minimizes log loss**.

The threshold is applied afterward. Logistic regression is not directly training to minimize FP and FN at a threshold of 0.50.